Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Quadratics

MARKSCHEME

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1. $x = 2, 3$ **M1A1** [3]
2. $x = -3, -4$ **M1A1** [3]
3. $x = 3, -\frac{1}{2}$ **M1A1** [4]
4. $x = \frac{2}{3}, -1$ **M1A1** [4]
5. $(x+3)^2 - 4$ **M1A1** [3]
6. $(x-4)^2 - 13$ **M1A1** [3]
7. $2(x+2)^2 - 9$ **M1A1** [4]
8. Min $= 3$ at $x = 2$ **M1A1** [3]
9. Max $= 6$ at $x = -1$ **M1A1** [4]
10. $x = 2 \pm \sqrt{5}$ **M1A1** [3]
11. $x = \frac{-3 \pm \sqrt{17}}{4}$ **M1A1** [4]
12. Intercepts $(-3, 0), (2, 0), (0, -6)$ **M1A1** [4]
13. Intercepts $(-2, 0), (4, 0), (0, -8)$; vertex $(1, -9)$ **M1A1** [5]
14. $\Delta = 1$; two distinct real roots **M1A1** [3]
15. $\Delta = -16$; no real roots **M1A1** [3]
16. $k^2 \geq 16 \Rightarrow |k| \geq 4$ **M1A1** [4]
17. $9 - 4k^2 < 0 \Rightarrow |k| > \frac{3}{2}$ **M1A1** [5]
18. $x = \pm 1, \pm 2$ **M1A1** [5]
19. $x = \pm 2, \pm 3$ **M1A1** [5]
20. $k = 1$ **M1A1** [3]
21. $x = -5 \pm \sqrt{7}$ **M1A1** [4]
22. $(3, -11)$ **M1A1** [4]
23. $x = 1, 3$ **M1A1** [5]
24. $x = -1, 2$ **M1A1** [5]
25. $(x+2)^2 + 1 \geq 1 > 0$ **M1A1** [4]